

HW 1

Townsend 1.3 (5 pts)

In Problem 3.2 we will see that the state of a spin- $\frac{1}{2}$ particle that is spin up along the axis whose direction is specified by the unit vector in spherical coordinates

$$\mathbf{n} = \sin \theta \cos \phi \mathbf{i} + \sin \theta \sin \phi \mathbf{j} + \cos \theta \mathbf{k}$$

with θ and ϕ being the familiar spherical coordinates shown in Fig. 1.11, is given by

$$|+\mathbf{n}\rangle = \cos \frac{\theta}{2} |+\mathbf{z}\rangle + e^{i\phi} \sin \frac{\theta}{2} |-\mathbf{z}\rangle$$

(a) Verify that the state $|+\mathbf{n}\rangle$ reduces to the states $|+\mathbf{x}\rangle$ and $|+\mathbf{y}\rangle$ given in the chapter for the appropriate choice of the angles θ and ϕ .

(b) Suppose that a measurement of S_z is carried out on a particle in the state $|+\mathbf{n}\rangle$. What is the probability that the measurement yields (i) $\hbar/2$? (ii) $-\hbar/2$?

(c) Determine the uncertainty ΔS_z of your measurements.

$$n = \sin \theta \cos \phi \hat{i} + \sin \theta \sin \phi \hat{j} + \cos \theta \hat{k}$$

A

The state $|+\mathbf{x}\rangle$ corresponds to the vector $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, and $|+\mathbf{y}\rangle$ to $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$.

To get only x , $\theta = \frac{\pi}{2}$ and $\phi = 0$.

$$|+\mathbf{x}\rangle = \cos \frac{\pi}{4} |+\mathbf{z}\rangle + \sin \frac{\pi}{4} |-\mathbf{z}\rangle$$

$$|+\mathbf{x}\rangle = \frac{\sqrt{2}}{2} (|+\mathbf{z}\rangle + |-\mathbf{z}\rangle)$$

Only y has $\theta = \frac{\pi}{2}$, $\phi = \frac{\pi}{2}$.

$$|+\mathbf{y}\rangle = \cos \frac{\pi}{4} |+\mathbf{z}\rangle + e^{i\frac{\pi}{2}} \sin \frac{\pi}{4} |-\mathbf{z}\rangle$$

$$\begin{aligned} |+\mathbf{y}\rangle &= \frac{\sqrt{2}}{2} (|+\mathbf{z}\rangle + i|-\mathbf{z}\rangle) \\ &= \sqrt{\frac{1}{2}} (|+\mathbf{z}\rangle + i|-\mathbf{z}\rangle) \end{aligned}$$

B

(b) Suppose that a measurement of S_z is carried out on a particle in the state $|+\mathbf{n}\rangle$. What is the probability that the measurement yields (i) $\hbar/2$? (ii) $-\hbar/2$?

$$|+\mathbf{n}\rangle = \cos \frac{\theta}{2} |+\mathbf{z}\rangle + e^{i\phi} \sin \frac{\theta}{2} |-\mathbf{z}\rangle$$

$$\begin{aligned} P\left(\frac{\hbar}{2}\right) &= \cos^2 \frac{\theta}{2} \\ P\left(-\frac{\hbar}{2}\right) &= \sin^2 \left(\frac{\theta}{2}\right) \end{aligned}$$

C

(c) Determine the uncertainty ΔS_z of your measurements.

$$|+n\rangle = \cos \frac{\theta}{2} |+z\rangle + e^{i\phi} \sin \frac{\theta}{2} |-z\rangle$$

$$\Delta S_z = \sqrt{\langle S_z^2 \rangle - \langle S_z \rangle^2}$$

We can find $\langle S_z \rangle^2$

$$\begin{aligned}\langle S_z \rangle &= \frac{\hbar}{2} (|\langle +z| + n \rangle|^2 - |\langle -z| + n \rangle|^2) \\ \langle S_z \rangle &= \frac{\hbar}{2} \left(\cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2} \right) \\ \langle S_z \rangle^2 &= \frac{\hbar^2}{4} \left(\cos^4 \frac{\theta}{2} + \sin^4 \frac{\theta}{2} - \cos^2 \frac{\theta}{2} \sin^2 \frac{\theta}{2} \right)\end{aligned}$$

We can also find $\langle S_z^2 \rangle$

$$\begin{aligned}\langle S_z^2 \rangle &= |\langle +z| + n \rangle|^2 \left(\frac{\hbar}{2} \right)^2 + \left(\frac{\hbar}{2} \right)^2 |\langle -z| + n \rangle|^2 \\ &= \frac{\hbar^2}{4} \left(\cos^2 \frac{\theta}{2} + \sin^2 \frac{\theta}{2} \right)\end{aligned}$$

This gives us

$$\begin{aligned}\Delta S_z &= \frac{\hbar}{2} \sqrt{\cos^2 \frac{\theta}{2} + \sin^2 \frac{\theta}{2} - \left(\cos^4 \frac{\theta}{2} + \sin^4 \frac{\theta}{2} - \cos^2 \frac{\theta}{2} \sin^2 \frac{\theta}{2} \right)} \\ &= \frac{\hbar}{2} \sqrt{1 - \left(\cos^4 \frac{\theta}{2} + \sin^4 \frac{\theta}{2} - \cos^2 \frac{\theta}{2} \sin^2 \frac{\theta}{2} \right)}\end{aligned}$$

I don't want to further simplify this, but I am sure there exist trig identities to do so.

Townsend 1.5 (5 pts)

- (a) What is the amplitude to find a particle that is in the state $|+n\rangle$ (from Problem 1.3) with $S_y = \hbar/2$? What is the probability? Check your result by evaluating the probability for an appropriate choice of the angles θ and ϕ .
- (b) What is the amplitude to find a particle that is in the state $|+y\rangle$ with $S_n = \hbar/2$? What is the probability?

$$\begin{aligned}|+y\rangle &= \sqrt{\frac{1}{2}} |+z\rangle + e^{\frac{i\pi}{2}} \sqrt{\frac{1}{2}} |-z\rangle \\ \Rightarrow \langle +y| &= \sqrt{\frac{1}{2}} \langle +z| + e^{-\frac{i\pi}{2}} \sqrt{\frac{1}{2}} \langle -z| \\ \langle +n| + y \rangle &= \left(\cos \frac{\theta}{2} \langle +z| + e^{i\phi} \sin \frac{\theta}{2} \langle -z| \right) \left(\sqrt{\frac{1}{2}} \langle +z| + e^{-\frac{i\pi}{2}} \sqrt{\frac{1}{2}} \langle -z| \right) \\ &\text{because } \langle +z| - z \rangle = 0 \\ \langle +n| + y \rangle &= \sqrt{\frac{1}{2}} \cos \frac{\theta}{2} + e^{i(\phi - \frac{\pi}{2})} \sqrt{\frac{1}{2}} \sin \frac{\theta}{2}\end{aligned}$$

The probability of this is

$$|\langle +z| + y \rangle|^2 = \frac{1}{2} \sin^2 \frac{\theta}{2} + \frac{1}{2} \cos^2 \frac{\theta}{2} + \frac{1}{2} \sin \frac{\theta}{2} \cos \frac{\theta}{2} \\ = \frac{1}{2} \left(1 + \sin \frac{\theta}{2} \cos \frac{\theta}{2} \right)$$

We can check this with the test case state $|+n\rangle = |+y\rangle$ to see if we get $\langle +y| + y \rangle = 1$.

$$\phi = \frac{\pi}{2}, \theta = \frac{\pi}{2}$$

This becomes

$$\sqrt{\frac{1}{2}} \sqrt{\frac{1}{2}} + \sqrt{\frac{1}{2}} \sqrt{\frac{1}{2}} \\ = 1$$

So the equality holds for the test case.

b)

$$|+n\rangle = \cos \frac{\theta}{2} |+z\rangle + e^{i\phi} \sin \frac{\theta}{2} |-z\rangle$$

We can write

$$\langle +y| + n \rangle \\ = \left(\sqrt{\frac{1}{2}} \langle +z| + e^{\frac{-i\pi}{2}} \sqrt{\frac{1}{2}} \langle -z| \right) \left(\cos \frac{\theta}{2} |+z\rangle + e^{i\phi} \sin \frac{\theta}{2} |-z\rangle \right) \\ \text{because } \langle +z| -z \rangle = 0 \\ \langle +y| + n \rangle = \sqrt{\frac{1}{2}} \cos \frac{\theta}{2} + e^{i(\phi - \frac{\pi}{2})} \sqrt{\frac{1}{2}} \sin \frac{\theta}{2}$$

This is the amplitude of getting $|+y\rangle$ given $|+n\rangle$. The probability is

$$|\langle +y| + n \rangle|^2 = \frac{1}{2} \left(1 + \sin \frac{\theta}{2} \cos \frac{\theta}{2} \right)$$

We can check this with $|+n\rangle = |+y\rangle$ to see if we get $\langle +y| + y \rangle = 1$.

$$\phi = \frac{\pi}{2}, \theta = \frac{\pi}{2}$$

This becomes

$$\sqrt{\frac{1}{2}} \sqrt{\frac{1}{2}} + \sqrt{\frac{1}{2}} \sqrt{\frac{1}{2}} \\ = 1$$

So the equality holds for the test case.

b

(b) What is the amplitude to find a particle that is in the state $|+y\rangle$ with $S_n = \hbar/2$? What is the probability?

$$|+n\rangle = \cos \frac{\theta}{2} |+z\rangle + e^{i\phi} \sin \frac{\theta}{2} |-z\rangle$$

$$|+y\rangle = \sqrt{\frac{1}{2}} |+z\rangle + e^{\frac{i\pi}{2}} \sqrt{\frac{1}{2}} |-z\rangle$$

This is exactly the same as pt a, since multiplication is commutative and the conjugate transpose acts on the same product:

$$\begin{aligned}
\langle +n | +y \rangle &= \left(\cos \frac{\theta}{2} | +z \rangle + e^{-i\phi} \sin \frac{\theta}{2} | -z \rangle \right) \left(\sqrt{\frac{1}{2}} \langle +z | + e^{i\frac{\pi}{2}} \sqrt{\frac{1}{2}} \langle -z | \right) \\
&\text{because } \langle +z | -z \rangle = 0 \\
\langle +n | +y \rangle &= \sqrt{\frac{1}{2}} \cos \frac{\theta}{2} + e^{i(\phi - \frac{\pi}{2})} \sqrt{\frac{1}{2}} \sin \frac{\theta}{2}
\end{aligned}$$

The probability is

$$\frac{1}{2} \left(1 + \cos \frac{\theta}{2} \sin \frac{\theta}{2} \right)$$

Townsend 1.6+ (5 pts)

Show that the state

$$|-n\rangle = \sin \frac{\theta}{2} | +z \rangle - e^{i\phi} \cos \frac{\theta}{2} | -z \rangle$$

- (a) Really does correspond to $|+\mathbf{n}\rangle$ with $\mathbf{n}$ pointing in the opposite direction.
- (b) Is normalized such that $\langle -n | -n \rangle = 1$.
- (c) Is orthogonal to $|+\mathbf{n}\rangle$ such that $\langle +n | -n \rangle = 0$ and $\langle -n | +n \rangle = 0$.
- (d) Show that as 3-vectors, $(-\mathbf{n}) \cdot \mathbf{n} = -1$.

Note that (d) is consistent with having the quantum the states be orthogonal: $\langle -n | +n \rangle = 0$. The vector $\mathbf{n}$ lives in physical 3D space whereas $|\mathbf{n}\rangle$ lives in a 2D abstract "Hilbert" space. The states $|\mathbf{n}\rangle$ and $|\mathbf{-n}\rangle$ are orthogonal because they give two different, definite, and mutually exclusive results when their spin is measured along the $\mathbf{n}$ direction.

a)

$$|+\mathbf{n}\rangle = \cos \frac{\theta}{2} | +z \rangle + e^{i\phi} \sin \frac{\theta}{2} | -z \rangle$$

We can check this by rotating only θ (the angle down from $+z$) by $\frac{\pi}{2}$. Any change to ϕ would not reflect the vector properly.

We see that $\cos \theta = \sin \theta + \frac{\pi}{2}$, so this holds.

b)

$$\langle -n | = \sin \frac{\theta}{2} \langle +z | - e^{-i\phi} \cos \frac{\theta}{2} \langle -z |$$

We can check normalization

$$\begin{aligned}
\langle -n | -n \rangle &= \left(\sin \frac{\theta}{2} \langle +z | - e^{-i\phi} \cos \frac{\theta}{2} \langle -z | \right) \left(\sin \frac{\theta}{2} | +z \rangle - e^{i\phi} \cos \frac{\theta}{2} | -z \rangle \right) \\
&= \sin^2 \frac{\theta}{2} + e^{2i\phi} \cos^2 \frac{\theta}{2} \\
&= \sin^2 \omega + \cos^2 \omega \\
&= 1
\end{aligned}$$

So this is properly normalized.

c)

$$\begin{aligned}
\langle -n | &= \sin \frac{\theta}{2} \langle +z | - e^{-i\phi} \cos \frac{\theta}{2} \langle -z | \\
| +n \rangle &= \cos \frac{\theta}{2} | +z \rangle + e^{i\phi} \sin \frac{\theta}{2} | -z \rangle
\end{aligned}$$

We can check orthogonality:

$$\begin{aligned}
 \langle -n | +n \rangle &= \left(\sin \frac{\theta}{2} \cos \frac{\theta}{2} \right) - \left(\cos \frac{\theta}{2} \sin \frac{\theta}{2} \right) \\
 &= 0
 \end{aligned}$$

So orthogonality holds

d)

Written component wise as a three vector,

$$n = \sin \theta \cos \phi \hat{i} + \sin \theta \sin \phi \hat{j} + \cos \theta \hat{k}$$

We can check the dot product $-n \cdot n$ by multiplying component wise (square each term, multiply each term by -1):

$$\begin{aligned}
 -n \cdot n &= -(\sin \theta \cos \phi)^2 - \sin^2 \theta \sin^2 \phi - \cos^2 \theta \\
 &= -\sin^2 \theta (\cos^2 \phi + \sin^2 \phi) - \cos^2 \theta \\
 &= -\sin^2 \theta (1) - \cos^2 \theta \\
 &= -(\sin^2 \theta + \cos^2 \theta) \\
 &= -1
 \end{aligned}$$

Thus, their orthogonality in Hilbert space has been shown.

Townsend 1.7 (5 pts)

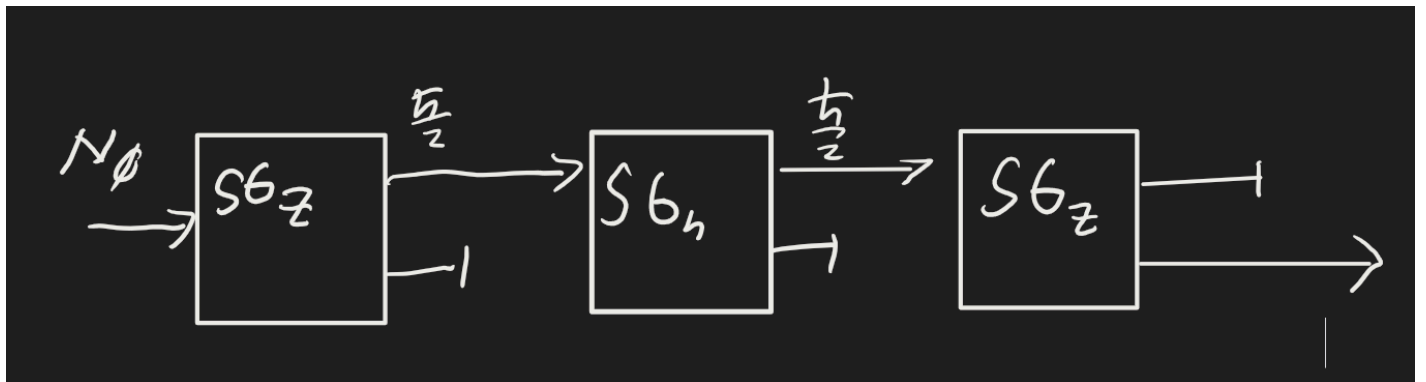
A beam of spin- $\frac{1}{2}$ particles is sent through a series of three Stern-Gerlach measuring devices, as illustrated in Fig. 1.12. The first SG_z device transmits particles with $S_z = \hbar/2$ and filters out particles with $S_z = -\hbar/2$. The second device, an SG_n device, transmits particles with $S_n = \hbar/2$ and filters out particles with $S_n = -\hbar/2$, where the axis **n** marks an angle θ in the x - z plane with respect to the z axis. Thus particles after passage through this SG_n device are in the state $|+n\rangle$ given in Problem 1.3 with the angle $\phi = 0$. A last SG_z device transmits particles with $S_z = -\hbar/2$ and filters out particles with $S_z = \hbar/2$.

- What fraction of the particles transmitted by the first SG_z device will survive the third measurement?
- How must the angle θ of the SG_n device be oriented so as to maximize the number of particles that are transmitted by the final SG_z device? What fraction of the particles survive the third measurement for this value of θ ?
- What fraction of the particles survive the last measurement if the SG_n device is simply removed from the experiment?

a)

Because $\phi = 0$, the states $|+n\rangle$ can be written succinctly as

$$|+n\rangle = \cos \frac{\theta}{2} |+z\rangle + \sin \frac{\theta}{2} |-z\rangle$$



The number of particles in the $|+n\rangle$ state after the second SG device is

$$\frac{N_0}{2} \langle +n | +z \rangle$$

This is

$$\langle +n | +z \rangle = \cos \frac{\theta}{2}$$

We can now multiply this with the probability $\langle -z | +n \rangle$

Which is $\sin \frac{\theta}{2}$. Therefore, the fraction of particles measured in the $-\frac{\hbar}{2}$ state after the third SG device is simply

$$\frac{1}{2} \cos \frac{\theta}{2} \sin \frac{\theta}{2}$$

B)

The angle which maximizes the function

$\cos \frac{\theta}{2} \sin \frac{\theta}{2}$ is $\frac{\pi}{2}$.

$$\cos \frac{\pi}{4} \sin \frac{\pi}{4} = 0.5$$

For this angle, there would be 0.25 particles after the third SG device (half through the first, then $\sqrt{\frac{1}{2}}$ for the next two).

C)

If the SG_n device is removed, then we are measuring in the same basis twice in a row. The particles do not "reset" their spin measurements.

We are asking the question $\langle -z | +z \rangle$ which is 0, because the two bases are orthogonal.

Townsend 1.15 (5 pts)

It is known that there is a 90% probability of obtaining $S_z = \hbar/2$ if a measurement of S_z is carried out on a spin- $\frac{1}{2}$ particle. In addition, it is known that there is a 20% probability of obtaining $S_y = \hbar/2$ if a measurement of S_y is carried out. Determine the spin state of the particle as completely as possible from this information. What is the probability of obtaining $S_x = \hbar/2$ if a measurement of S_x is carried out?

We can write out the information as

$$\begin{aligned} |\langle +z | +n \rangle|^2 &= \frac{9}{10} \\ |\langle +y | +n \rangle|^2 &= \frac{1}{5} \end{aligned}$$

This becomes a system of equations, and we can solve for $|+n\rangle$.

$$|+n\rangle = \cos \frac{\theta}{2} |+z\rangle + e^{i\phi} \sin \frac{\theta}{2} |-z\rangle$$

$$\langle +z | +n \rangle = \frac{9}{10} = \cos^2 \frac{\theta}{2}$$

$$2 \arccos\left(\sqrt{\frac{9}{10}}\right) = \theta$$

$$\theta \approx 0.6435$$

Similarly,

$$|\langle +y|+n\rangle|^2 = \left(\sqrt{\frac{1}{2}}\langle +z| + \sqrt{\frac{1}{2}}\langle -z| \right) \left(\cos \frac{\theta}{2} | +z\rangle + e^{i\phi} \sin \frac{\theta}{2} | -z\rangle \right)^2$$

$$\frac{1}{5} = \left(\sqrt{\frac{1}{2}} \cos \frac{\theta}{2} + \sqrt{\frac{1}{2}} e^{i\phi} \sin \frac{\theta}{2} \right)^2$$

$$\frac{1}{\sqrt{5}} = \sqrt{\frac{1}{2}} (\cos 0.32 + \sin(0.32)e^{i\phi})$$

$$-1 = \frac{\sqrt{\frac{2}{5} - \cos(0.32)}}{\sin(0.32)} = e^{i\phi}$$

$$\phi = \frac{\pi}{2}$$

$$|+n\rangle = \cos(0.32)|+z\rangle + e^{i\frac{\pi}{2}} \sin(0.32)|-z\rangle$$

Because $\phi \approx \frac{\pi}{2}$, $\langle +z|+n\rangle \approx \frac{1}{2}$.

Townsend 2.8 (5 pts)

The column vector representing the state $|\psi\rangle$ is given by

$$|\psi\rangle \xrightarrow{S_z \text{ basis}} \frac{1}{\sqrt{5}} \begin{pmatrix} i \\ 2 \end{pmatrix}$$

Using matrix mechanics, show that $|\psi\rangle$ is properly normalized and calculate the probability that a measurement of S_x yields $\hbar/2$. Also determine the probability that a measurement of S_y yields $\hbar/2$.

Normalized?

We can check

$$\langle \psi | \psi \rangle \stackrel{?}{=} \frac{1}{5} [-i, 2] \begin{bmatrix} i \\ 2 \end{bmatrix} = \frac{1}{5} 1 + 4 = 1$$

We can see that this is indeed normalized.

$\frac{\hbar}{2}$?

$$|x\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$|+y\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

$$|\psi\rangle = \frac{1}{\sqrt{5}} \begin{bmatrix} i \\ 2 \end{bmatrix}$$

Probability of getting Y as $\frac{\hbar}{2}$

$$|\langle +y | \psi \rangle|^2 = \frac{1}{\sqrt{2}} (1, -i) \frac{1}{\sqrt{5}} \begin{pmatrix} i \\ 2 \end{pmatrix}^2$$

$$= \frac{1}{\sqrt{10}} (i - 2i) \frac{1}{\sqrt{10}} (-i + 2i)$$

$$= \frac{1}{10}$$

Probability of getting X as $\frac{\hbar}{2}$

$$\begin{aligned}(\langle +x|\psi\rangle)^2 &= \sqrt{\frac{1}{2}} \begin{bmatrix} 1 & 1 \end{bmatrix} \frac{1}{\sqrt{5}} \begin{bmatrix} i \\ 2 \end{bmatrix}^2 \\&= \sqrt{\frac{1}{10}} (i+2) \\&= \frac{1}{10} (i+2)(-i+2) \\&= \frac{1}{10} 5 \\&= \frac{1}{2}\end{aligned}$$